

UBT705: GENOMICS AND PROTEOMICS

L	T	P	Cr
3	1	0	3.5

Course Objective: The objective of this course is to familiarize Students with concepts of genomics and proteomics technologies and to understand the steps of high throughput data analysis.

Syllabus

Introduction: Goals, applications and limitations of Bioinformatics, Biological sequence and molecule file formats, Biological databases. Types of biological databases, its classification, application and pitfalls. Information retrieval from biological databases, pairwise sequence alignment and database searching: Evolutionary Basis of sequence alignment, Homologous sequence, Global alignment and local alignment, Gap penalties, Scoring matrices, Methods of sequence alignment e.g., Dynamic programming methods: Needleman-Wunsch and Smith-Waterman algorithm, Database similarity search, Heuristic methods: FASTA and BLAST.

Genomics and Transcriptomics: Structure of Gene and genomes of Eukaryotic and prokaryotic, genome databases, Sanger sequencing-principle, methodology and applications, Whole genome - de novo sequencing or resequencing; exome sequencing, RNA sequencing; small RNA sequencing; Next Generation Sequencing (NGS) workflow, Differential expression.

NGS Data Analysis: NGS work flow and pipeline for data analysis. Next generation sequence data analyses, Data format, Quality control-Phred score; FastQC and FastX tool kits, read length, read depth, sequence coverage, Homology (orthology groups), Genome alignment and analysis tools- BWA (BurrowsWheeler Aligner), SAMtools, GATK (The Genome Analysis Toolkit), Cuffcompare, Velvet, Oases, Trinity.

Proteomics: Introduction to quantitative proteomics- Differential proteomics, post-translational modifications, Proteogenomic Concepts and principles of genome annotation, genome search specific peptides, alternative translation initiation, small ORFs, Analysis of transcriptomic and proteomic data for genome annotation; Gene prediction algorithms.

Structural Bioinformatics: Protein structure basics, Visualization, Classification. Protein secondary structure prediction. Protein tertiary structure prediction: methods, applications, RNA structures, Genomics and proteomics.

Laboratory Work (if applicable)

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Course Learning Objectives (CLO)

The students will be able to:

1. Explore biological databases
2. Perform pre-processing of NGS data
3. Analyze the algorithms to handle NGS data
4. Comprehend various methods involved in data analysis of genomics and proteomics technology for commercial and industrial applications.

Text Books

1. Xiong J, Essential Bioinformatics, Cambridge University Press (2013).
2. S B Primrose and R Twyman, Principles of gene manipulation and Genomics (2013) 7th Edition.
3. Mount D W, Bioinformatics - Sequence and Genome Analysis, Cold Spring Harbour Laboratory Press (2001), 2nd ed.
4. Ghosh Z, and Mallick B, Bioinformatics – Principles and Applications, Oxford University Press (2010).

Reference Books

1. Higgins, D. and Taylor, W., Bioinformatics: Sequence, Structure and Databanks – A Practical Approach, Oxford University Press (2010).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments /quizzes)	30

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